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Comparative Effectiveness of Sleeve Gastrectomy and Semaglutide for Weight Loss and Metabolic Outcomes: A Prospective Non-Randomized Study

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Abstract

Background: Bariatric surgery and pharmacotherapy represent two major strategies for the treatment of obesity. Sleeve gastrectomy (SG) is the most widely performed surgical procedure and is supported by long-term evidence, while semaglutide, a glucagon-like peptide-1 receptor agonist (GLP-1 RA), has recently emerged as an effective medical therapy. However, prospective real-world studies directly comparing these interventions are limited.

Methods: We conducted a prospective, non-randomized trial at a single tertiary center (ChiCTR2300070632). Adults with obesity were allocated to SG ($n = 36$) or once-weekly semaglutide ($n = 35$) based on shared decision-making. Semaglutide was administered for a planned duration of 24 weeks, after which continuation or discontinuation was determined by patient preference. Primary outcomes were percentage total weight loss (%TWL) and changes in body mass index (BMI) at 1, 3, 6, and 12 months. Secondary outcomes included changes in metabolic parameters.

Results: SG achieved significantly greater weight loss at all time points. At 12 months, mean %TWL was 28.6% for SG versus 11.3% for semaglutide ($p < 0.001$). Mean BMI decreased from 32.7 ± 2.8 to 23.8 ± 2.6 kg/m² in the SG group and from 32.0 ± 2.6 to 28.4 ± 4.0 kg/m² in the semaglutide group ($p < 0.001$). Both groups demonstrated metabolic improvements, but SG was associated with greater reductions in triglycerides and greater increases in high-density lipoprotein cholesterol (HDL-C). Among patients who discontinued semaglutide after 6 months ($n = 25$), mean weight increased from 74.6 ± 11.3 kg to 83.2 ± 12.3 kg by 12 months, with 32% regaining to baseline weight or higher.

Conclusions: SG achieved greater weight loss and improvements in triglycerides and HDL-C compared with semaglutide. Semaglutide was effective during treatment, but many patients experienced weight regain after discontinuation.

Trial registration: Trial Registry No. ChiCTR2300070632 (2023 April 19th)

Keywords: Obesity; Sleeve gastrectomy; Semaglutide; Weight loss; Pharmacotherapy

Background

Obesity is a chronic, relapsing, and multifactorial disease that contributes substantially to global morbidity and mortality [1-3]. It is closely associated with type 2 diabetes mellitus (T2DM), dyslipidemia, cardiovascular disease, and certain cancers, and its prevalence continues to rise worldwide [1-3]. As the health and economic burden of obesity escalates, the development of effective and sustainable treatment strategies has become a pressing priority for healthcare systems.

Metabolic and bariatric surgery (MBS) has long been established as the most effective intervention for substantial and durable weight loss, with consistent evidence of improvements in cardiometabolic outcomes and long-term survival [4-7]. Sleeve gastrectomy (SG), in particular, has become the most widely performed surgical procedure due to its relative technical simplicity, favorable safety profile, and robust short- and mid-term outcomes [4]. Multiple international and national guidelines have been developed to standardize patient selection, perioperative care, and follow-up for MBS, reflecting decades of accumulated experience and high-level evidence [8-9].

In parallel, advances in pharmacotherapy have opened a new era for medical treatment of obesity. Glucagon-like peptide-1 receptor agonists (GLP-1 RAs), such as semaglutide, have demonstrated clinically meaningful weight loss and improvements in metabolic parameters in randomized controlled trials [10, 11]. In many patients, pharmacotherapy provides a non-invasive alternative or complement to surgery, and its uptake has accelerated in routine practice.

Despite the rapid adoption of semaglutide and the established role of SG, there remains limited prospective, real-world evidence directly comparing the two approaches. Existing studies typically focus on either surgical or pharmacological interventions in isolation, leaving unanswered questions regarding relative effectiveness, sustainability, and metabolic benefits when assessed side by side [12-14].

The present prospective, non-randomized study was conducted to compare SG with semaglutide in adults with obesity, focusing on weight loss, metabolic outcomes, and treatment sustainability over a 12-month follow-up period. By directly evaluating these two leading strategies, we aim to provide timely evidence to guide individualized treatment selection and inform the design of future guidelines for obesity management.

Methods

Study Design and Registration

This was a prospective, non-randomized controlled trial registered in the Chinese Clinical Trial Registry (ChiCTR2300070632). The study was conducted at a single tertiary referral center, and all patients provided written informed consent prior to participation. Ethical approval was obtained from the institutional review board, and the study was performed in accordance with the Declaration of Helsinki.

Participants

Between April, 2023 and August, 2023, 80 consecutive patients with obesity were screened for eligibility. Inclusion criteria followed Chinese guidelines for bariatric and metabolic surgery, which recommend intervention for patients with a body mass index (BMI) ≥32.5 kg/m² irrespective of comorbidities, or ≥27.5 kg/m² in the presence of T2DM or other obesity-related metabolic complications [15]. Patients with suspected thyroid malignancy, history of pancreatitis, contraindications to semaglutide, or contraindications to surgery were excluded. Eligible patients were enrolled consecutively within a predefined BMI range (30–35 kg/m²).

Of the 80 patients enrolled, 40 elected to undergo laparoscopic SG and 40 chose semaglutide therapy. Treatment allocation was determined through shared decision-making after standardized counseling regarding efficacy, risks, costs, and follow-up requirements, without physician-directed assignment.

Figure 1 summarizes patient enrollment, allocation, and follow-up. Ultimately, 36 patients in the laparoscopic SG group and 35 in the semaglutide group completed the 6-month follow-up and were included in the per-protocol analysis.

Laparoscopic Sleeve Gastrectomy

All operations were performed by the same experienced surgical team using a standardized technique. A 36 Fr bougie was employed for calibration, and the gastric resection extended from approximately 4 cm proximal to the pylorus to the angle of His. Staple line reinforcement was routinely applied by suturing. Standard perioperative care and postoperative dietary progression were followed according to institutional protocols.

Semaglutide Therapy

Patients assigned to pharmacotherapy received once-weekly subcutaneous semaglutide injections in the abdominal wall at a fixed time each week. The dose escalation regimen was based on the Semaglutide Treatment Effect in People with obesity (STEP) 6 trial protocol, which is tailored for East Asian populations. Treatment began at 0.25 mg weekly for 2 weeks, followed by 0.5 mg for 2 weeks, 1.0 mg for 2 weeks, and escalation to a maintenance dose of 1.5 mg weekly [16]. The planned treatment duration was 24 weeks. After this period, continuation or discontinuation of semaglutide was determined by patient preference and tolerability, reflecting real-world clinical practice. Patients also received standardized lifestyle counseling: dietitians recommended a low-calorie balanced diet with optional 16:8 intermittent fasting, and rehabilitation specialists encouraged at least 150 minutes of moderate-intensity physical activity per week (e.g., brisk walking, swimming, elliptical training) [17, 18].

Outcomes and Follow-up

Patients were assessed at baseline, and at 1, 3, 6, and 12 months. Primary outcomes included percent total weight loss (%TWL) and changes in BMI. Secondary outcomes included changes in metabolic markers (glycated hemoglobin A1c/HbA1c, lipid profile, liver function, uric acid), and the incidence of gastroesophageal reflux disease

(GERD) by GERD-Questionnaire (GERD-Q) [19]. For the semaglutide group, outcomes were further stratified according to whether patients discontinued or continued therapy beyond 6 months. This allowed assessment of both the planned treatment effect and the sustainability of weight loss after discontinuation.

Statistical Analysis

Continuous variables were expressed as means ± standard deviations and compared using Student's t test or Mann–Whitney U test as appropriate. A two-sided p value <0.05 was considered statistically significant. Analyses were performed on a per-protocol basis, including only patients who completed treatment and 12-month follow-up.

Results

Patient Enrollment and Follow-up

A total of 80 patients were enrolled between April and August 2023 (40 in the SG group and 40 in the semaglutide group). Figure 1 summarizes patient allocation and follow-up. In the semaglutide group, one patient was excluded due to suspected thyroid malignancy, two discontinued treatments due to intolerable gastrointestinal side effects, and two discontinued due to gout flares attributed to hyperuricemia, leaving 35 patients for analysis. In the SG group, four patients were lost to follow-up, leaving 36 for analysis.

Baseline Characteristics

Baseline demographic and clinical characteristics are shown in Table 1. The two groups were comparable in terms of age, sex distribution, height, weight, and BMI. However, patients in the SG group had significantly higher HbA1c (6.3% vs 5.5%, $p = 0.011$) and triglyceride levels (1.94 vs 1.45 mmol/L, $p = 0.024$) at baseline compared with the semaglutide group. Other parameters, including total cholesterol, HDL-C, low-density lipoprotein cholesterol (LDL-C), liver enzymes, and uric acid, were similar between groups.

Weight Loss Outcomes and GERD

Weight loss outcomes are summarized in Table 2. At one month, the SG group achieved a mean %TWL of 11.1% compared with 6.3% in the semaglutide group ($p < 0.001$). This difference persisted at 3 months (19.0% vs 13.2%, $p < 0.001$) and 6 months (24.9% vs 18.5%, $p < 0.001$). By 12 months, SG patients had maintained a significantly greater weight reduction, with a mean %TWL of 28.6% compared with 11.3% in the semaglutide group ($p < 0.001$). The 12-month semaglutide data included both patients who discontinued treatment after 6 months (n = 25) and those who continued therapy beyond 6 months (n = 10); subgroup outcomes are presented in Table 4.

Absolute body weight and BMI trends mirrored these findings. SG patients experienced a progressive reduction in mean body weight from 89.9 ± 14.4 kg at baseline to 64.6 ± 10.2 kg at 12 months, whereas semaglutide patients reduced from

90.1 ± 11.8 kg to 80.9 ± 12.3 kg ($p < 0.001$). Mean BMI decreased from 32.7 ± 2.8 to 23.8 ± 2.6 kg/m² in the SG group, compared with a decrease from 32.0 ± 2.6 to 28.4 ± 4.0 kg/m² in the semaglutide group at 12 months ($p < 0.001$).

At 6 months, gastroesophageal reflux symptoms were assessed using the GERD-Q questionnaire. A score ≥8 was reported in 16 of 36 SG patients (44.4%) compared with 7 of 35 semaglutide patients (20.0%). Although this difference did not reach statistical significance ($p = 0.070$), it suggested a trend toward higher GERD symptoms in the SG group. It should be noted that GERD-Q was only taken after treatment, and baseline GERD status was not available for comparison.

Laboratory Outcomes

Changes in laboratory parameters at 6 months are presented in Table 3. Both groups showed improvements in glycemic and lipid profiles compared with baseline. Between-group comparison demonstrated that SG patients achieved significantly greater improvements in triglycerides (Δ0.94 ± 0.87 vs 0.40 ± 0.71 mmol/L, $p = 0.005$) and HDL-C (Δ+0.32 ± 0.29 vs +0.05 ± 0.32 mmol/L, $p < 0.001$). Differences in HbA1c reduction, total cholesterol, LDL-C, liver enzymes, albumin, and uric acid did not reach statistical significance.

Weight Trajectory After Semaglutide Discontinuation

Table 4 summarizes outcomes among patients who discontinued semaglutide after 6 months (n = 25). In this subgroup, mean body weight decreased from 90.7 ± 12.1 kg at baseline to 74.6 ± 11.3 kg at 6 months but rebounded to 83.2 ± 12.3 kg by 12 months. Eight patients (32%) regained to baseline weight or higher, while 6 patients (24%) maintained 5–9.9% weight loss, and 11 patients (44%) maintained >10% weight loss at 12 months. Reasons for semaglutide discontinuation are summarized in Table 5. Documented causes included gastrointestinal adverse effects, treatment cost, pregnancy planning, and patient-perceived sufficiency of weight loss, while a proportion discontinued for undocumented reasons.

Discussion

In this prospective non-randomized study, we compared the effectiveness of SG and once-weekly semaglutide therapy in patients with obesity. Our findings demonstrate that SG achieved greater and more durable weight loss over 12 months, with significantly larger reductions in BMI and %TWL compared with semaglutide. SG was also associated with greater improvements in triglyceride and HDL-C levels, while other metabolic parameters improved similarly in both groups. At the same time, we observed that semaglutide induced clinically meaningful weight loss in the first 6 months, consistent with results from the STEP trials, but weight regain occurred in many patients after treatment discontinuation [12, 14, 16]. Importantly, a subset of patients who continued semaglutide beyond 6 months maintained greater weight reduction, underscoring the relevance of treatment persistence.

The superiority of bariatric surgery in achieving long-term weight loss and metabolic improvements has been consistently demonstrated across randomized and observational studies [20-22]. Our findings reinforce this established evidence, showing that SG patients maintained close to 30% TWL at one year.

On the other hand, pharmacotherapy with GLP-1 receptor agonists represents a newer but rapidly expanding approach. Large randomized controlled trials such as STEP 1 and STEP 6 have reported weight losses of 10–15% after 68 weeks of semaglutide treatment, confirming its clinical efficacy [16, 23]. In our study, semaglutide patients achieved an average TWL of 18.5% at 6 months, which is consistent with these prior reports. However, when many patients discontinued treatment at 6 months, the mean TWL decreased to 11.3% at 12 months. This pattern mirrors other real-world reports showing that discontinuation of semaglutide often leads to partial or full weight regain, emphasizing the chronic nature of obesity and the pharmacologic principle that treatment efficacy is contingent upon sustained use [24]. The reasons for semaglutide discontinuation in our cohort further illustrate the complexity of real-world pharmacotherapy adherence. In our study, discontinuation was driven by a heterogeneous set of factors, including gastrointestinal adverse effects, treatment cost, pregnancy planning, and patient-perceived sufficiency of weight loss. Importantly, discontinuation was not exclusively attributable to negative experiences, as a subset of patients elected to stop therapy after achieving weight loss outcomes they considered satisfactory at 6 months.

Surgery and pharmacotherapy should not be viewed as competing interventions but as complementary components of a multimodal strategy. Bariatric surgery remains the most effective treatment for achieving substantial and durable weight loss, especially in patients with severe obesity or obesity-related complications [5-7]. However, semaglutide offers a valuable alternative for patients who are unwilling or ineligible for surgery, and it may serve as a bridge to surgery or as an adjunct in cases of weight regain.

Recent large-scale observational and trial-based studies have highlighted that incretin-based therapies, including dual Glucose-dependent insulinotropic polypeptide / glucagon-like peptide-1 receptor agonists (GIP/GLP-1 receptor agonists) such as tirzepatide, may confer benefits beyond weight loss, including reductions in all-cause mortality, major adverse cardiovascular events, and renal outcomes [25]. However, such clinical outcome studies involve substantially larger populations and longer follow-up periods than the present investigation. Accordingly, our study was designed to focus on prospective, real-world comparisons of weight loss trajectories and metabolic surrogate outcomes over a 12-month period, which may represent intermediate markers relevant to cardiometabolic risk reduction.

Importantly, our findings underscore a major gap in obesity care: while bariatric surgery has mature guidelines governing indications, perioperative care, and follow-up, comparable frameworks for pharmacotherapy are lacking. Key unanswered questions include the optimal duration of therapy, criteria for discontinuation, strategies for monitoring after cessation, and protocols for re-initiating treatment in the event of weight regain. Without such guidance, clinical practice remains variable and patients may face inconsistent management.

Several research priorities emerge from our study. First, it remains unclear whether semaglutide and similar GLP-1 receptor agonists can or should be continued indefinitely. While obesity is a chronic disease that may warrant lifelong treatment, the long-term safety, tolerability, and cost of indefinite therapy are not yet established [26, 27]. Second, the effectiveness of restarting pharmacotherapy after discontinuation and weight regain requires further investigation. It is possible that retreatment may restore weight loss, but prospective studies are needed to confirm this. Third, the potential role of combination or sequential therapy deserves exploration. For example, semaglutide could be used preoperatively to reduce surgical risk, or postoperatively to maintain weight loss and reduce the likelihood of regain [28-30]. Such integrated strategies may ultimately provide more personalized and sustainable care.

Another dimension to be considered is economic. Bariatric surgery has been shown in numerous analyses to be cost-effective, largely due to long-term reductions in diabetes, cardiovascular disease, and other obesity-related complications [31, 32]. By contrast, semaglutide and related agents represent a significant ongoing expense, particularly when sustained for years or potentially for life [26, 27]. Cost-effectiveness analyses for pharmacotherapy remain limited and are highly sensitive to assumptions about treatment duration and adherence. As these drugs gain wider use, robust modeling studies will be essential to inform policy decisions, reimbursement frameworks, and clinical guidelines.

The strengths of this study include its prospective design, head-to-head comparison of two major obesity treatments, and real-world analysis of patients who either discontinued or continued semaglutide beyond the planned 6 months. These features provide valuable insights into the practical challenges of treatment adherence and sustainability. However, several limitations must be acknowledged. The study was non-randomized and conducted at a single center with a modest sample size, which limits generalizability and may introduce selection bias. Baseline differences in HbA1c and triglycerides between groups could have influenced outcomes, although the primary findings were robust. The follow-up period was limited to 12 months; longer follow-up is needed to assess durability of both surgical and pharmacological effects. Additionally, GERD-Q scores were collected only after treatment, and baseline data were not available. Therefore, changes in GERD symptoms from baseline could not be assessed, and our findings should be interpreted with caution. Socioeconomic factors and patient preferences, which may influence treatment choice and adherence, were not formally quantified and may represent unmeasured confounders inherent to the non-randomized design.

Conclusions

SG achieved superior weight loss and metabolic improvements compared with semaglutide in this prospective 12-month study, though it was associated with a higher incidence of GERD symptoms. Semaglutide therapy provided meaningful short-term weight loss but discontinuation frequently led to weight regain, emphasizing the importance of treatment persistence. Rather than viewing surgery and pharmacotherapy as competing approaches, they should be seen as complementary tools within a continuum of obesity care. Future guidelines must address critical gaps in the use of pharmacotherapy, including duration, discontinuation, and re-initiation strategies. Further research is needed to determine whether long-term or lifelong pharmacotherapy is feasible, how best to integrate drugs and surgery, and the cost-effectiveness of these approaches in diverse healthcare systems. Together, these efforts will help shape a more comprehensive and patient-centered framework for obesity management.

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Abbreviations

ALT	Alanine transaminase
AST	Aspartate transaminase
BMI	Body mass index
GERD	Gastroesophageal reflux disease
GERD-Q	GERD-Questionnaire
GLP-1 RA	Glucagon-like peptide-1 receptor agonist
GIP/GLP-1 receptor agonists	Glucose-dependent insulinotropic polypeptide / glucagon-like peptide-1 receptor agonists
HbA1c	Glycated hemoglobin A1c
HDL-C	High-density lipoprotein cholesterol
LDL-C	Low-density lipoprotein cholesterol
LSG	Laparoscopic sleeve gastrectomy
MBS	Metabolic and bariatric surgery
SG	Sleeve gastrectomy
%TWL	Percentage of total weight loss
T2DM	Type 2 diabetes mellitus

Declarations

Acknowledgments

The authors have no acknowledgements to declare.

Consent for Publication

Not applicable. This study does not contain any individual person's data in any form (including individual details, images, or videos).

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Data Availability

The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Authors' Contributions

YY and QZ contributed equally to this work. YY, QZ, SY, YP, KS, XS1, XS2, QW, RZ, GZ, and ZC contributed to study design, data collection, data analysis, and manuscript drafting. JW, HC, and YC contributed to study conception, critical revision of the manuscript, and overall supervision. All authors read and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Ethics Approval and Consent to Participate

This study was approved by the Institutional Review Board of our institution (approval number: 2023.133). This study was registered at the Chinese Clinical Trial Registry (ChiCTR2300070632) (https://www.chictr.org.cn/showproj.html?proj=192488). Informed consent was obtained from all individual participants included in the study.

Figure Legend

Figure 1. Figure 1. CONSORT flow diagram of patient enrolment, allocation, follow-up, and analysis.

A total of 80 patients were initially enrolled and randomized to receive either Semaglutide (n = 40) or sleeve gastrectomy (SG; n = 40). One patient was excluded prior to treatment due to suspected thyroid malignancy. In the Semaglutide group, four patients discontinued treatment because of intolerable gastrointestinal reactions (n = 2) or hyperuricemia-induced gout pain (n = 2), leaving 35 patients for final analysis. In the SG group, no patients discontinued therapy; however, four patients were lost to follow-up, resulting in 36 patients available for final analysis at 12 months.

Table Legend

Table 1. Baseline Characteristics. Values are means±SD. *P* values are for the overall comparisons. *SG* sleeve gastrectomy, *BMI* body mass index, *HbA1C* Glycated hemoglobin A1c, *HDL-C* high-density lipoprotein cholesterol, *LDL-C* low-density lipoprotein cholesterol, *AST* aspartate transaminase, *ALT* alanine transaminase, *GERD* gastroesophageal reflux disease.

Table 2. Weight loss and GERD outcomes during follow-up. Values are means±SD. *P* values are for the overall comparisons. *SG* sleeve gastrectomy, *%TWL* percent of total weight loss, *BMI* body mass index, *GERD* gastroesophageal reflux disease. * Includes patients who continue and stopped medications after 6 months of treatment.

Table 3. Changes in laboratory values after 6-month treatment. Values are means±SD. *P* values are for the overall comparisons. Δ reduction in the variable, *SG* sleeve gastrectomy, *HbA1C* glycosylated hemoglobin, *HDL-C* high-density lipoprotein cholesterol, *LDL-C* low-density lipoprotein cholesterol, *AST* aspartate transaminase, *ALT* alanine transaminase.

Table 4. Weight outcomes in patients who discontinued semaglutide after 6-month treatment. Values are means±SD. *P* values are for the overall comparisons. Δ reduction in the variable, *SG* sleeve gastrectomy, *HbA1C* glycosylated hemoglobin, *HDL-C* high-density lipoprotein cholesterol, *LDL-C* low-density lipoprotein cholesterol, *AST* aspartate transaminase, *ALT* alanine transaminase.

Table 5. Reasons for discontinuation of semaglutide therapy after 6 months of treatment. Values are presented as number (percentage) of patients.

Table 1. Baseline Characteristics.

Variable	**SG (N = 36)**	**Semaglutide (N = 35)**	***P* value**
Age (years)	33.0 ± 7.0	33.7 ± 7.1	0.7082
Female, *n* (%)	25 (69.4%)	19 (54.3%)	0.4173
Height (cm)	165.3 ± 8.9	167.5 ± 7.6	0.2573
Weight (kg)	89.9 ± 14.4	90.1 ± 11.8	0.9439
BMI (kg/m²)	32.7 ± 2.8	32.0 ± 2.6	0.3145
HbA1C (%)	6.3 ± 1.6	5.5 ± 0.4	*0.0111*
Total Cholesterol, mmol/L	4.95 ± 0.91	4.62 ± 0.87	0.1171
Triglycerides, mmol/L	1.94 ± 0.92	1.45 ± 0.86	0.0238
HDL-C, mmol/L	1.12 ± 0.22	1.19 ± 0.28	0.2669
LDL-C, mmol/L	3.08 ± 0.76	2.96 ± 0.68	0.5081
Albumin, g/L	46.7 ± 3.0	47.5 ± 3.4	0.3058
ALT, U/L	42.1 ± 23.1	39.5 ± 31.4	0.6920
AST, U/L	26.4 ± 11.3	27.5 ± 16.7	0.7330
Uric Acid, μmol/L	419.5 ± 99.5	416.7 ± 100.3	0.9069

Values are means±SD. *P* values are for the overall comparisons. *SG* sleeve gastrectomy, *BMI* body mass index, *HbA1C* glycosylated hemoglobin, *HDL-C* high-density lipoprotein cholesterol, *LDL-C* low-density lipoprotein cholesterol, *ALT* aspartate transaminase, *ALT* alanine transaminase, *GERD* gastroesophageal reflux disease.

Table 2. Weight loss and GERD outcomes during follow-up.

Variable		SG (N = 36)	Semaglutide (N = 35)	*P* value
% TWL (%)	1 month	11.1 ± 2.2	6.3 ± 3.1	<0.001
	3 months	19.0 ± 3.3	13.2 ± 4.5	<0.001
	6 months	24.9 ± 5.3	18.5 ± 6.0	<0.001
	12 months	28.6 ± 7.8	11.3 ± 10.8*	<0.001
Weight (kg)	1 month	79.9 ± 12.2	84.4 ± 11.0	0.1038
	3 months	72.7 ± 11.0	78.2 ± 10.3	0.0336
	6 months	67.3 ± 10.6	73.4 ± 10.4	0.0171
	12 months	64.6 ± 10.2	80.9 ± 12.3*	<0.001
BMI (kg/m2)	1 month	29.0 ± 2.5	30.0 ± 2.4	0.1173
	3 months	26.5 ± 2.3	27.9 ± 2.4	0.0130
	6 months	24.5 ± 2.4	26.0 ± 20.6	0.0129
	12 months	23.8 ± 2.6	28.4 ± 4.0*	<0.001
GERD, n (%)	6 months	16 (44.4%)	7 (20.0%)	0.0704

Values are means±SD. *P* values are for the overall comparisons. *SG* sleeve gastrectomy, *%TWL* percent of total weight loss, *BMI* body mass index, *GERD* gastroesophageal reflux disease. * Includes patients who continue and stopped medications after 6 months of treatment.

Table 3. Changes in laboratory values after 6-month treatment.

Variable	**SG (N = 36)**	**Semaglutide (N = 35)**	***P* value**
ΔHbA1C (%)	0.8 ± 1.2	0.4 ± 0.4	0.0928
ΔTotal Cholesterol, mmol/L	0.39 ± 0.73	0.40 ± 0.62	0.9717
ΔTriglycerides, mmol/L	0.94 ± 0.87	0.40 ± 0.71	*0.0054*
ΔHDL-C, mmol/L	(+) 0.32 ± 0.29	(+) 0.05 ± 0.32	*0.0005*
ΔLDL-C, mmol/L	0.09 ± 0.78	0.02 ± 0.52	0.3169
ΔAlbumin, g/L	0.9 ± 2.8	1.6 ± 5.1	0.4844
ΔALT, U/L	22.6 ± 30.6	24.7 ± 29.8	0.7706
ΔAST, U/L	7.8 ± 10.7	10.9 ± 15.8	0.3432
ΔUric Acid, µmol/L	95.1 ± 88.0	84.6 ± 70.0	0.5811

Values are means±SD. *P* values are for the overall comparisons. Δ reduction in the variable, *SG* sleeve gastrectomy, *HbA1C* glycosylated hemoglobin, *HDL-C* high-density lipoprotein cholesterol, *LDL-C* low-density lipoprotein cholesterol, *ALT* aspartate transaminase, *ALT* alanine transaminase.

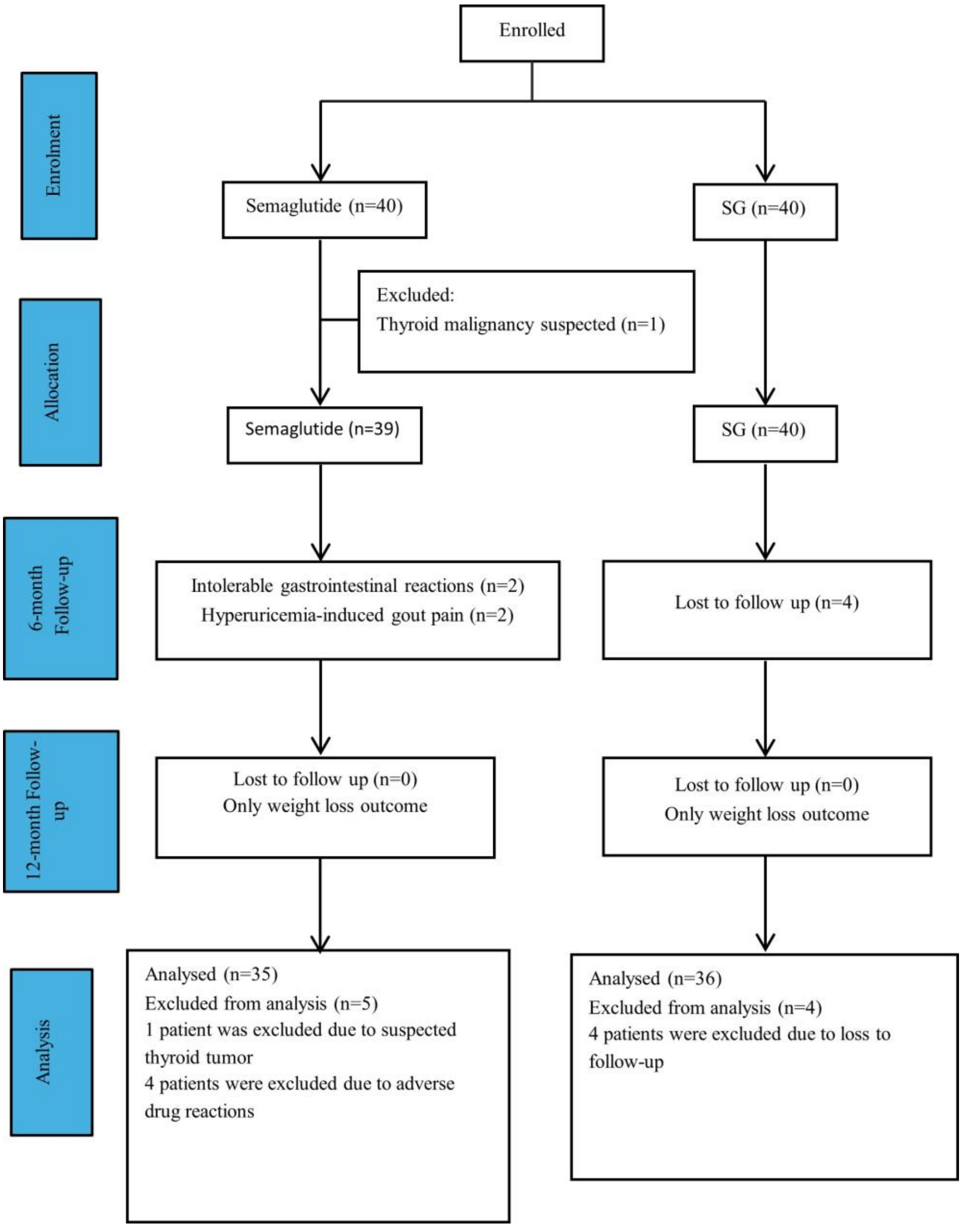
Table 4. Weight outcomes in patients who discontinued semaglutide after 6-month treatment.

Variable	Patients discontinued semaglutide after 6-month treatment (N = 25)
Baseline Weight, kg	90.7 ± 12.1
Weight at 6 months, kg (last use of Semaglutide)	74.6 ± 11.3
Weight at 12 months, kg (after 6 months of discontinuing Semaglutide)	83.2 ± 12.3
Regain to Baseline or More	8 (32%)
Maintained 5-9.9% Weight Loss	6 (24%)
Maintained >10% Weight Loss	11 (44%)

Values are means±SD. *P* values are for the overall comparisons.

Table 5. Reasons for discontinuation of semaglutide therapy after 6 months of treatment.

Reason for discontinuation	**n**	**%**	**Description**
High treatment cost	3	12	Drug is self-paid
Gastrointestinal side effects	4	16	Nausea, vomiting, abdominal discomfort, etc.
Satisfied with weight-loss outcome and perceived no need for continuation	6	24	Patients felt weight loss results were adequate
Pregnancy planning	3	12	-
Unknown reasons	9	36	-
Total discontinued	25	100%	-


Enrolled
Enrolment
Semaglutide (n=40)
SG (n=40)
Excluded:
Thyroid malignancy suspected (n=1)
Allocation
Semaglutide (n=39)
SG (n=40)
6-month Follow-up
Intolerable gastrointestinal reactions (n=2)
Hyperuricemia-induced gout pain (n=2)
Lost to follow up (n=4)
12-month Follow-up
Lost to follow up (n=0)
Only weight loss outcome
Lost to follow up (n=0)
Only weight loss outcome
Analysis
Analysed (n=35)
Excluded from analysis (n=5)
1 patient was excluded due to suspected thyroid tumor
4 patients were excluded due to adverse drug reactions
Analysed (n=36)
Excluded from analysis (n=4)
4 patients were excluded due to loss to follow-up